

Francis Bach: Learning Theory from First Principle

pinto

Chapter 1. Mathematical Preliminaries

Exercise (1.1). For $\alpha \in \mathbb{R}$ such that $\alpha \neq -1/n$ and $\mathbf{1}_n \in \mathbb{R}^n$ the vector of all 1s, show that we have $(I + \alpha \mathbf{1}_n \mathbf{1}_n^\top)^{-1} = I - \frac{\alpha}{1+n\alpha} \mathbf{1}_n \mathbf{1}_n^\top$.

Solution

Chapter 2. Introduction to Supervised Learning

Exercise (2.1). Consider binary classification with $\mathcal{Y} = \{-1, 1\}$ and the loss function

$$\ell(-1, -1) = \ell(1, 1) = 0, \quad \ell(-1, 1) = c_- > 0, \quad \ell(1, -1) = c_+ > 0,$$

where c_- is the cost of a false positive and c_+ is the cost of a false negative. Compute a Bayes predictor at x as a function of $\eta(x) := \mathbb{E}[y \mid x]$.

Solution

With $\ell(y, z) = c_+ \mathbf{1}(y = 1, z = -1) + c_- \mathbf{1}(y = -1, z = 1)$, we have

$$\begin{aligned} \mathcal{R}(f) &= \mathbb{E}[\ell(y, z)] \\ &= \mathbb{E}[c_+ \mathbf{1}(y = 1, z = -1) + c_- \mathbf{1}(y = -1, z = 1)] \\ &= \mathbb{E}[c_+ \mathbb{P}\{y = 1|x\} \mathbf{1}(z = -1) + c_- \mathbb{P}\{y = -1|x\} \mathbf{1}(z = 1)]. \end{aligned}$$

Observe that $\mathbb{E}[y|x] = \mathbb{P}\{y = 1|x\} - \mathbb{P}\{y = -1|x\}$ and $\mathbb{P}\{y = 1|x\} + \mathbb{P}\{y = -1|x\} = 1$. Then,

$$\mathcal{R}(f) = \mathbb{E}\left[c_+ \frac{1 + \mathbb{E}[y|x]}{2} \mathbf{1}(z = -1) + c_- \frac{1 - \mathbb{E}[y|x]}{2} \mathbf{1}(z = 1)\right]$$

The optimal classifier will select the most likely class weighted with the cost. Using the notation $\eta(x) = \mathbb{E}[y|x]$, then, if $c_+ \frac{1+\eta(x)}{2} - c_- \frac{1-\eta(x)}{2} > 0$, $f_*(x) = 1$, while if $c_+ \frac{1+\eta(x)}{2} - c_- \frac{1-\eta(x)}{2} < 0$, $f_*(x) = -1$.

Exercise (2.2). Consider a learning problem on $\mathcal{X} \times \mathcal{Y}$, with $\mathcal{Y} = \mathbb{R}$ and the absolute loss defined as $\ell(y, z) = |y - z|$. Compute a Bayes predictor $f_* : \mathcal{X} \rightarrow \mathbb{R}$.

Solution

With $\ell(y, z) = |y - z|$, we have

$$\begin{aligned}\mathcal{R}(f) &= \mathbb{E}_{(x,y)}[\ell(y, z)] \\ &= \mathbb{E}_x[\mathbb{E}_{y|x}[|y - z||x|]] \\ &= \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} |y - z| f(y|x) dy \right) f(x) dx \\ &= \int_{-\infty}^{\infty} \left(\int_{-\infty}^z (z - y) f(y|x) dy + \int_z^{\infty} (y - z) f(y|x) dy \right) f(x) dx.\end{aligned}$$

Observe that we want to minimize $\mathbb{E}_{y|x}[|y - z||x|]$.

$$\frac{\partial \mathbb{E}[|y - z||x|]}{\partial z} = \frac{\partial}{\partial z} \left(\int_{-\infty}^z (z - y) f(y|x) dy + \int_z^{\infty} (y - z) f(y|x) dy \right) = 0.$$

Using Leibniz's integral rule for each term, for $\int_{-\infty}^z (z - y) f(y|x) dy$ we have

$$\frac{\partial}{\partial z} \int_{-\infty}^z (z - y) f(y|x) dy = (z - z) f(z|x) \cdot \frac{d}{dz} z + \int_{-\infty}^z \frac{\partial}{\partial z} [(z - y) f(y|x)] dy = \int_{-\infty}^z f(y|x) dy.$$

Similarly, for $\int_z^{\infty} (y - z) f(y|x) dy$ we have

$$\frac{\partial}{\partial z} \int_z^{\infty} (y - z) f(y|x) dy = -(z - z) f(z|x) \cdot \frac{d}{dz} z + \int_z^{\infty} \frac{\partial}{\partial z} [(y - z) f(y|x)] dy = - \int_z^{\infty} f(y|x) dy.$$

The second term in the Leibniz's rule goes to 0 since we're differentiating a constant. This gives us $\int_{-\infty}^z f(y|x) dy = \int_z^{\infty} f(y|x) dy$. Thus, the Bayes predictor is $f_*(x) =$ the median of $y|x$.

Exercise (2.3). Consider a learning problem on $\mathcal{X} \times \mathcal{Y}$, with $\mathcal{Y} = \mathbb{R}$ and the "pinball" loss

$$\ell(y, z) = \alpha(y - z)_+ + (1 - \alpha)(z - y)_+,$$

for $\alpha \in (0, 1)$. Compute a Bayes predictor $f_* : \mathcal{X} \rightarrow \mathbb{R}$. Provide an interpretation in terms of quantiles.

Solution

With $\ell(y, z) = \alpha(y - z)_+ + (1 - \alpha)(z - y)_+$, we have

$$\begin{aligned}\mathcal{R}(f) &= \mathbb{E}_{(x,y)}[\ell(y, z)] \\ &= \mathbb{E}_x[\mathbb{E}_{y|x}[\ell(y, z)|x]] \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (\alpha(y - z)_+ + (1 - \alpha)(z - y)_+) f(y|x) dy f(x) dx \\ &= \int_{-\infty}^{\infty} \left(\int_{-\infty}^z (1 - \alpha)(z - y) f(y|x) dy + \int_z^{\infty} \alpha(y - z) f(y|x) dy \right) f(x) dx.\end{aligned}$$

Observe that we want to minimize $\mathbb{E}_{y|x}[\ell(y, z)|x]$.

$$\frac{\partial \mathbb{E}[\ell(y, z)|x]}{\partial z} = \frac{\partial}{\partial z} \left(\int_{-\infty}^z (1 - \alpha)(z - y) f(y|x) dy + \int_z^{\infty} \alpha(y - z) f(y|x) dy \right) = 0.$$

Using Leibniz's integral rule, same as Exercise 2.2, we have

$$(1 - \alpha) \int_{-\infty}^z f(y|x) dy = \alpha \int_z^{\infty} f(y|x) dy \rightarrow \int_{-\infty}^z f(y|x) dy = \alpha$$

Thus, the Bayes predictor is $f_*(x) =$ the 100α th quantiles of the distribution $y|x$. A more definitive way to represent $f_*(x)$ is $F_{y|x}^{-1}(\alpha)$ or $\min\{z : F(z) \geq \alpha\}$ in the case where F is not invertible.

Exercise (2.4). Characterize Bayes predictors for regression with the “ ε -insensitive” loss defined as $\ell(y, z) = \max\{0, |y - z| - \varepsilon\}$. If for each x, y is supported in an interval of length less than 2ε , what are the Bayes predictors?

Solution

We have

$$\begin{aligned}\mathcal{R}(f) &= \mathbb{E}[\ell(y, x)] \\ &= \mathbb{E}[\max\{0, |y - z| - \varepsilon\}] \\ &= \mathbb{E}_x[\mathbb{E}_{y|x}[\cdot]] \\ &= \mathbb{E}\end{aligned}$$

Exercise (2.5). (Inverting predictions) Consider the binary classification problem with $\mathcal{Y} = \{-1, 1\}$ and the 0-1 loss. Relate the risk of a prediction f to that of its opposite $-f$.

Exercise (2.6). (“Chance” predictions) Consider binary classification problems with the 0-1 loss. What is the risk of a random prediction rule where we predict the two classes with equal probabilities independent of input x ? Address the same question with multiple categories.

Exercise (2.7). Consider a random prediction rule where we predict from the probability distribution of y given x . When is this achieving the Bayes risk?